

1. Student Management system

```
class student
```

```
    PRIVATE name, rollNO, marks
```

```
    METHOD acceptDetails()
```

```
        INPUT name, roll NO, marks
```

```
    END METHOD
```

```
    METHOD calculateAverage()
```

```
        RETURN marks/ subjects
```

```
    END METHOD
```

```
    METHOD displayResult()
```

```
        IF calculateAverage() >= 40
```

```
            PRINT "Pass"
```

```
        ELSE
```

```
            PRINT "Fail"
```

```
        END METHOD
```

```
END class
```

2. LIBRARY BOOK MANAGEMENT

```
class Book
```

```
    PRIVATE title, author, ISBN, is_Available
```

```
    METHOD issue_BOOK()
```

```
        is_Available = FALSE
```

```
    END METHOD
```

```
    METHOD returnBook()
```

```
        is_Available = TRUE
```

```
    END METHOD
```

```
    METHOD displayDetail()
```

```
        PRINT title, author, ISBN, is_Available
```

```
    END METHOD
```

```
END class
```


3. Employee Salary Calculation

```
class Employee
    Private EMPID, Name, basicPay
    Method calculateGross()
        Return basicPay + HRA + DA
    End Method
    Method calculateNet()
        Return calculateGross() - PF - Tax
    End Method
    Method displayPaySlip()
        PRINT calculateNet()
    End Method
End Class
```

4. Shape Area Calculator

```
Class Shape
    Method calculateArea()
End Class

Class Circle Extends Shape
    Method calculateArea()
        Return  $3.14 * r * r$ 
    End Method
End Class

Class Rectangle Extends Shape
    Method calculateArea()
        Return  $l * b$ 
    End Method
End Class
```

5. Vehicle Registration System

```
Class Vehicle
    regNo, owner, fuelType
    Method displayDetail()
End Class
```



```
class TwoWheeler extends vehicle  
    engine capacity
```

```
END CLASS
```

G BANK Account Hierarchy

```
class BankAccount  
    acc NO, holder, balance
```

```
END CLASS
```

```
class Savings Account extends BankAccount
```

```
    Method withdraw(amount)
```

```
        If amount <= balance
```

```
            balance = balance - amount
```

```
END CLASS
```

```
class Current Account extends BankAccount
```

```
    Method withdraw(amount)
```

```
        If amount <= balance + overdraft
```

```
            balance = balance - amount
```

```
END CLASS
```

7. Hospital Record System.

```
class Patient
```

```
    Private diagnosis, prescription
```

```
    Method generate Bill()
```

```
END CLASS
```

```
class Inpatient extends Patient
```

```
    Method generate Bill()
```

```
        Return days * daily charge
```

```
END CLASS
```


8. E-commerce Product catalog

```
class Product
    price
    method applyDiscount()
END CLASS

class Electronics extends Product
    method applyDiscount()
        return price * 0.90
    end class

class Clothing extends Product
    method applyDiscount()
        return price * 0.80
    end class

class Grocery extends Product
    method applyDiscount()
        return price * 0.95
    end class
```

9. University staff Hierarchy

```
class Staff
    private salary
    protected name
    method calculateAllowance()
end class

class TeachingStaff extends Staff
end class

class Professor extends TeachingStaff
end class
```



```
calculate lecturer extends TeachingStaff  
end class
```

```
class nonTeachingStaff extends Staff  
end class
```

10.

Smart Home Device Controller

```
class SmartDevice
```

```
  method executeCommand(cmd)
```

```
end class
```

```
class SmartLight extends SmartDevice
```

```
  method executeCommand(cmd)
```

```
    print "Light Action"
```

```
end class
```

```
class SmartThermostat extends SmartDevice
```

```
  method executeCommand(cmd)
```

```
    print "Temperature Action"
```

```
end class
```

```
class SmartLock extends SmartDevice
```

```
  private pin
```

```
  method executeCommand(cmd)
```

```
    print "Lock Action"
```

```
end class
```